

Supporting Information for

In Vivo Efficacy and Biochemical Target Engagement of a Chromene-based Inhibitor of Trypanosoma cruzi Lysyl-tRNA Synthetase

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Table S1. Crystallographic data collection and refinement statistics.

	<i>TcKRSc-Lys</i>	<i>TcKRSc-5-Lys</i>
Wavelength	0.9772	0.9772
Resolution range	47.45 - 2.08 (2.13 - 2.08)	47.33 - 2.8 (3.02 - 2.8)
Space group	P 43 21 2	P 43 21 2
Unit cell (x, y, z)	59.97, 59.97, 310.3	60.1 60.1 307.177
(α, β, γ)	90, 90, 90	90 90 90
Total reflections	893993 (65717)	205109 (35324)
Unique reflections	35668 (2646)	16747 (3243)
Multiplicity	25.1 (24.8)	12.2 (10.9)
Completeness (%)	99.8 (97.5)	96.87 (84.44)
Mean I/sigma(I)	14.3 (1.5)	5.99 (0.85)
Wilson B-factor	47.8	95.4
R-merge	0.156 (2.999)	0.2915 (2.998)
R-meas	0.159 (3.061)	0.3041 (3.145)
R-pim	0.043 (0.842)	0.08589 (0.9403)
CC1/2	0.998 (0.456)	0.996 (0.398)
Reflections used in refinement	33768	14393
Free reflections (%)	5	5
R-work	0.19	0.26
R-free	0.25	0.31
Number of non-hydrogen atoms	4031	3594
macromolecules	3955	3566
ligands	10	25
solvent	66	3
Protein residues	461	490
RMS(bonds)	0.93	2
RMS(angles)	1.41	0.44
Ramachandran		
 favored (%)	93	94
 allowed (%)	6	6
 outliers (%)	0.6	0.2
Rotamer outliers (%)	3.73	1.60
Clashscore	7.74	3.87
Average B-factor (all atoms)	63.00	117.47
 macromolecules	63.75	117.70
 ligands	45	90.55
 solvent	48.9	70.58
PDB ID	9YJN	9YJR

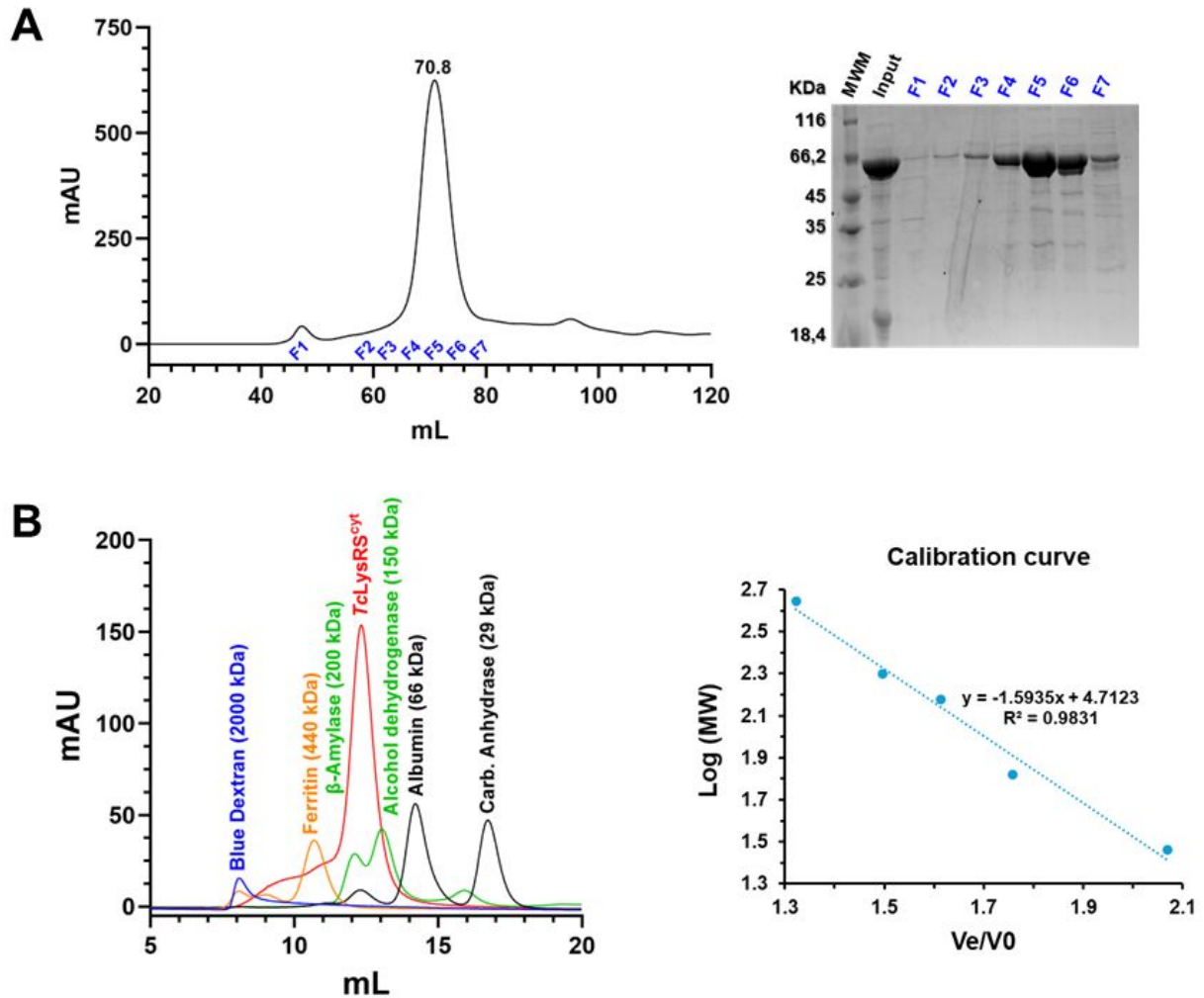


Figure S1. Purity and homogeneity of recombinant cytosolic lysyl-tRNA synthetase from *Trypanosoma cruzi* (*TcLysRS^{cyt}*). **A)** Representative preparative size-exclusion chromatography (SEC) profile and SDS-PAGE analysis of *TcLysRS^{cyt}* samples obtained during the final purification steps. **B)** Analytical SEC analysis showing that full-length *TcLysRS^{cyt}* (red line) eluted as a predominant oligomeric species with an apparent molecular weight of approximately 190 ± 20 kDa. This apparent mass exceeds the theoretical dimeric mass (~ 132 kDa), likely reflecting the hydrodynamic behavior of the protein in SEC, but supports that *TcLysRS^{cyt}* is not monomeric in solution. Molecular weight standards are indicated in the chromatogram; each line represents an independent run.

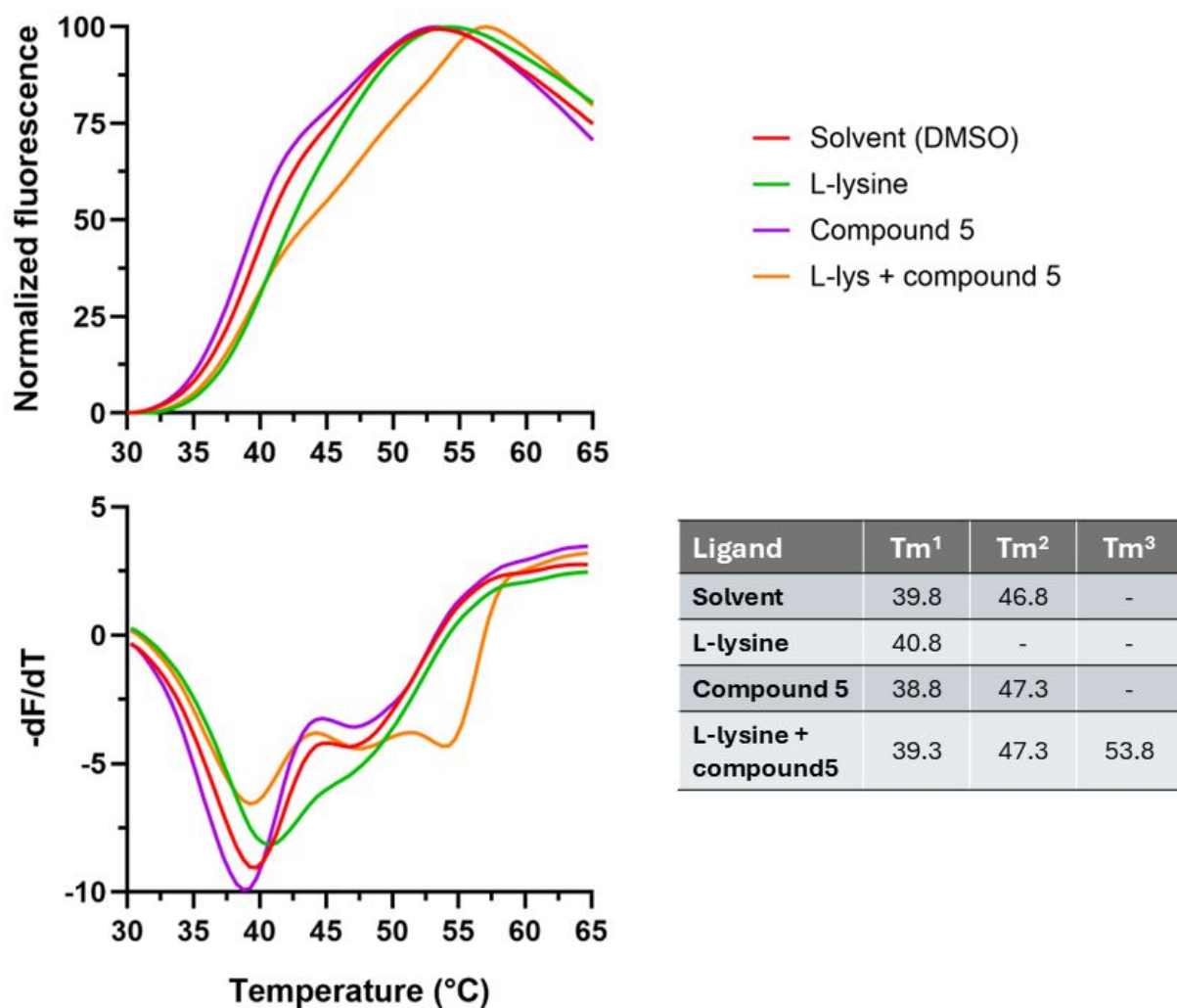


Figure S2. Differential scanning fluorimetry analysis of *TcLysRS^{cyt}* in the absence and presence of ligands. Thermal shift denaturation curves show that *TcLysRS^{cyt}* displays a biphasic unfolding profile both in the apo condition and in the presence of individual ligands, such as lysine or compound 5. In contrast, the combined presence of lysine and compound 5 produces an additional transition phase, suggesting that simultaneous ligand binding stabilizes a distinct conformational or population state of the enzyme.

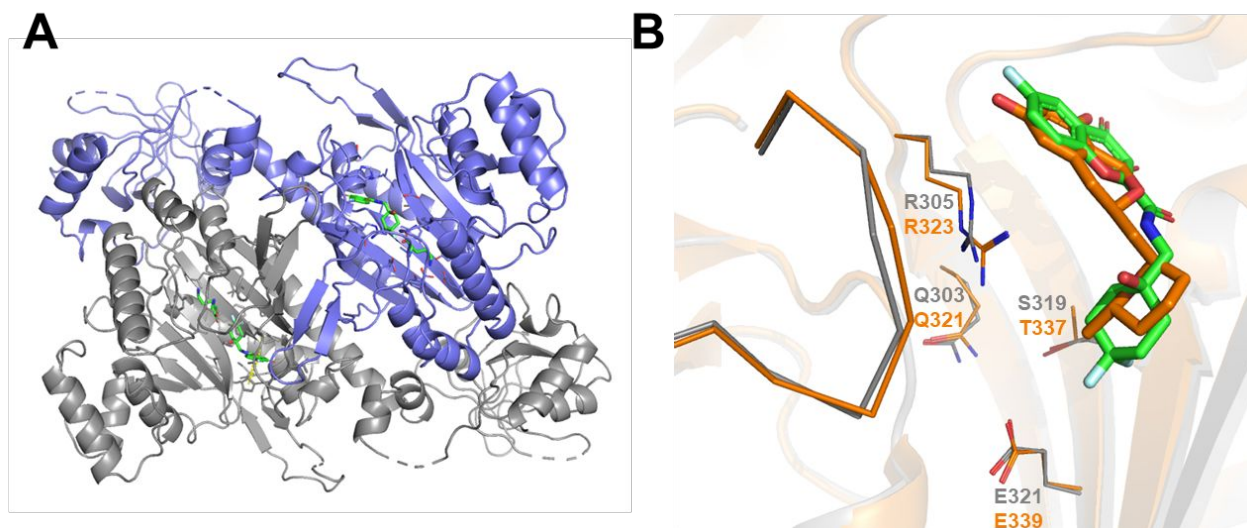


Figure S3. Structure of *Trypanosoma cruzi* lysyl-tRNA synthetase (*TcLysRS^{cyt}*). **A)** Overall *TcLysRS* structure and visualization of the dimer (with two protomers, one depicted in blue and the other as gray), reconstituted through crystallographic symmetry; the active site contains both lysine and compound 5, illustrated as green sticks. **B)** Structural alignment of *TcLysRS* (gray) and human LysRS (*HsLysRS*; orange), each complexed with compound 5 (sticks with green carbons) and cladosporin (sticks with orange carbons), respectively, demonstrate key residues involved in stabilizing an adjacent loop. These residues are conserved between human and *T. cruzi* LysRS, but differ in *Plasmodium* and *C. parvum* LysRS, which has importance for compound selectivity.

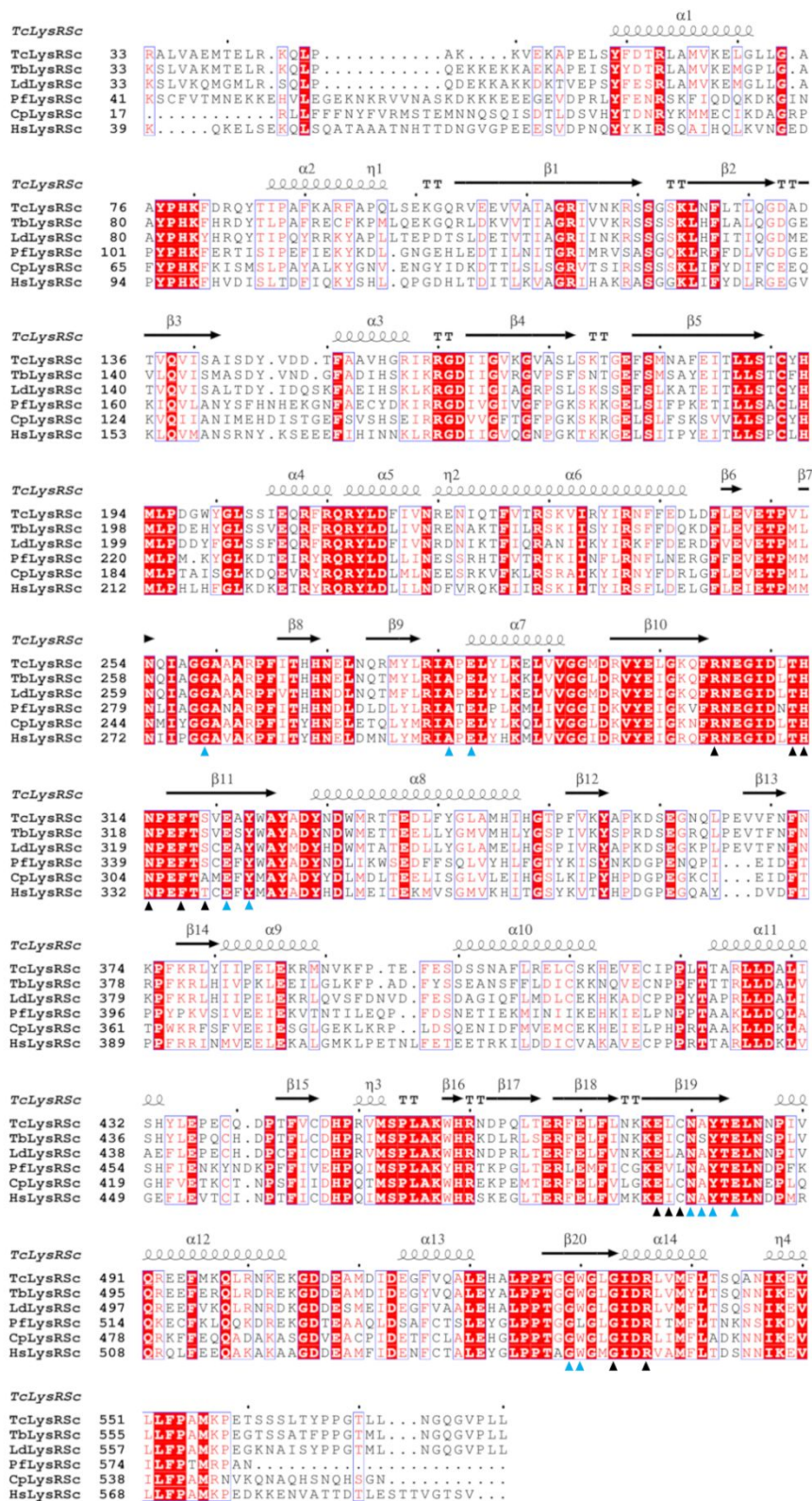


Figure S4. Sequence alignment of cytosolic lysyl-tRNA synthetases (LysRS) from human and parasitic organisms. Residues forming the ATP- (black arrowheads) and L-lysine-binding pockets (blue arrowheads) are indicated. Compound 5 is an inhibitor that binds to the ATP pocket. Secondary-structure elements of *TcLysRS*^{cyt} are shown above the aligned sequences. The sequence alignment was generated using Clustal Omega¹ and rendered with ESPrnt 3.2.² Abbreviations: **Tc**, *T. cruzi*, **Tb**, *T. brucei*, **Ld**, *L. donovani*, **Pf**, *P. falciparum*, **Cp**, *C. parvum*, **Hs**, *H. sapiens*.

Supplementary References

- (1) Madeira, F.; Madhusoodanan, N.; Lee, J.; et al. The EMBL-EBI Job Dispatcher sequence analysis tools framework in 2024. *Nucleic Acids Research*. **2024**, 52(W1):W521-W525. DOI: 10.1093/nar/gkae241.
- (2) Gouet, P.; Robert, X.; and Courcelle, E. ESPript/ENDscript: extracting and rendering sequence and 3D information from atomic structures of proteins. *Nucl. Acids Res.* **2003**, 31(13), 3320-3323. doi: 10.1093/nar/gkg556.